Smile-adjusted delta hedging for options improved with boosting

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Abstract

The Black-Scholes delta ratio does not minimize variance of value changes in positions including options and their underlying assets because it assumes a constant volatility for underlying price changes. Hull and White (2017) proposed an empirical model that adjusts the Black-Scholes delta into the minimum variance delta. They considered the expected change in the implied volatility as a quadratic function of the Black-Scholes delta given an underlying price change. The model reduced hedging variance for options on the S&P500 index and individual stocks between 2004 and 2015, but the performance depended on a relationship between the underlying price change and its volatility. This study investigates how statistical learning using regression trees can improve the delta adjustment and hedging performance from the Hull and White model.

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Chapter 1

Introduction

1.1 Motivations

The seminal work of Black and Scholes (1973) and Merton (1973) has set a common practice in options pricing and hedging. The model assumes constant volatility in underlying price changes. "Practitioner Black-Scholes delta" or the BS delta is calculated from the model as the partial derivative of the option price with respect to its underlying price given other variables kept constant, including the volatility. The volatility can be reverse-engineered from the observed option price with the Black-Scholes formula and the resulting number is called " implied volatility".

However, the implied volatility is observed to be not constant in practice. Therefore, the BS delta does not provide a hedging strategy that minimizes variance in hedging residuals, and "practitioner Black-Scholes vega" or the BS vega must be also considered. The BS vega is the partial derivative of the option price with respect to volatility given other variables kept constant.

1.2 Context of the Study

Hull and White (2017) investigated an empirical model to estimate adjustments to the BS delta to minimize variance in hedging residuals. The resulting ratio is called the minimum variance delta. Not only does it take into account the underlying price change as the BS delta does, it also considers the expected change in volatility conditional on the underlying price change. They assume that (1) option price depends only on the underlying price and its implied volatility, (2) these two variables follow diffusion processes, and (3) exponentials of the underlying price changes are negligible. Given other variables kept constant, volatility smile is a pattern of the implied volatility with respect to the underlying price change. Hence, their development of delta ratio can be termed as smile-adjusted delta.

The Hull and White (2017) model yielded a good performance in out-of-the-money call options on the S&P 500 index between 2004 and 2015. Nonetheless, the gain became worse for in-the-money options. The limited effectiveness was mainly due to more noise in the relationship between implied volatility changes and underlying price changes, which we note due to the above assumptions.

1.3 Objectives and Contributions

Within the aforementioned context, the first objective of this thesis is to examine how the empirical model performs for options on the S&P 500 index between 2015 and 2017. The outcome would indicate the model consistency across a different timeframe and with a shorter in-sample period.

The second objective is to develop an additional boosting step using regression trees and improve the adjustment to the BS delta by studying on hedging residuals of the empirical model. This method is intuitive and potential as it leverages feature selection capacity of the boosting method to weigh contributions of inputs and optimizes the forecast function. Thus, the proposed method is expected to overcome limitations of the assumptions in the original model that relies solely on underlying price changes and the expected volatility changes.

1.4 Overview of the Thesis

After Chapter 2 revisits the Black-Scholes framework, Chapter 3 presents the concept of volatility smile to highlight the relationship between volatility changes and underlying price changes. Chapter 4 introduces supervised learning including gradient boosting regression trees.

Chapter 5 provides a primary idea how Hull and White (2017) came up with their empirical model estimating the minimum variance delta. We also apply their model on data covered in this study. In such context, the chapter proposes the boosting framework that is supposed to improve on the original model.

Chapter 6 shows empirical results for options on the S&P 500 index in 2017.

Chapter 7 presents conclusions.

Chapter 2

The Black-Scholes model

It is straightforward that a trader is supposed to buy a call option if they think its underlying price will increase higher than its strike price by at least its price without knowing any valuation model. However, a model is needed when we want to describe and compare options with different maturities, underlying assets and strikes at any given time.

Black and Scholes (1973) and Merton (1973) developed a model known as the Black-Scholes-Merton (BSM) model or Black-Scholes (BS) model that was a major breakthrough in pricing of European stock options. Black and Scholes (1973) used the capital asset pricing model to determine the elasticity of the option price with respect to the stock price. Merton's approach was different by setting up a riskless portfolio consisting of an option and its underlying stock. Then, he argued the portfolio return over a short period of time must be risk-free rate. As a result, Merton's approach depended on weaker assumptions.

A model that perfectly captures the real world is difficult to have because financial markets are too complicated to calibrate. Thus, the BS model also has certain simplifying assumptions and its range of applicability that traders should be aware of. In addition, there have been many research papers attempting to relax the assumptions by adjustment methods, and to work on other different models for better option pricing and hedging.

2.1 The BS assumptions

Stochastic underlying price process

The model is set in a continuous-time financial market with a risky asset S_t follows the dynamics:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \tag{2.1}$$

where μ is constant drift and σ is constant volatility of S_t; W_t is a standard Wiener process.

Constant risk-free rate

The risk-free rate r is constant in all time horizons, and equal for borrowing and lending.

Frictionless market

The model is set in a financial market without transaction costs or taxes. All securities are also perfectly divisible.

Short selling and no arbitrage opportunity

The model market allows short selling and there is no arbitrage opportunity.

2.2 The BS formula for European plain vanilla options pricing

The famous BSM pricing formulas for European call and put options respectively are

$$\begin{aligned} C_t^{BS} &= S_t e^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2) \\ P_t^{BS} &= Ke^{-r\tau}\Phi(-d_2) - S_t e^{-q\tau}\Phi(-d_1) \end{aligned} \tag{2.2}$$

as

$$\begin{aligned} d_1 &= \frac{\log(S_t/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \\ d_2 &= d_1 - \sigma\sqrt{T} \end{aligned} \tag{2.3}$$

where T is time to maturity in years, S_t is underlying price at time t, K is option strike price, q is stock dividend yield, r is the risk-free rate, and $\Phi(x)$ is the cumulative probability distribution function for a variable with a standard normal distribution.

Definition 2.1 *The cp flag denotes a binary variable that equals to 1 for a call and 0 for a put option.*

Eventually, the BSM formula becomes as

$$BS_t(S_t,\ \sigma,\ cp\ flag,\ K,\ T,\ r,\ q) = C_t^{BS}\mathbb{1}_{cp\ flag=1} + P_t^{BS}\mathbb{1}_{cp\ flag=0} \tag{2.4}$$

2.3 The Greeks

The Greeks measure the sensitivity of option prices to a small change in underlying parameters. They are important in risk management. The most common of the Greeks are delta, gamma, vega, theta and rho. Given f is the option price and $\phi(x)$ is the probability density function for a variable with a standard normal distribution, formulas of the Greeks based on the BS model are presented as below.

Delta is the most important Greek in risk management. Delta measures the rate of change of option value with respect to changes in its underlying price $\delta = \frac{\partial f}{\partial S}$, which is equal to

$$\begin{aligned} \delta_{BS_{\text{call}}} &= e^{-qT}\Phi(d_1) \\ \delta_{BS_{\text{put}}} &= -e^{-qT}\Phi(-d_1) \end{aligned} \tag{2.5}$$

respectively for call and put options based on the BS model.

For a vanilla option, delta is between 0 and 1 for a call option, and between -1 and 0 for a put option. Also, the BS delta is commonly considered as a proxy for probability that the option will expire in-the-money.

Remark 2.1 Apart from a ratio between S and K, the BS delta $\delta_{BS} = \partial f / \partial S$ is also related to moneyness of options.

Remark 2.2 Delta is presented as an instantaneous percentage of the total number of underlying shares represented by one option contract in a delta-neutral portfolio.

Gamma measures the rate of change of delta with respect to changes in its underlying price $\gamma = \frac{\partial \delta}{\partial S} = \frac{\partial^2 f}{\partial S^2}$, which is equal to

$$\gamma_{BS} = e^{-qT} \frac{\phi(d_1)}{S\sigma\sqrt{T}} = Ke^{-rT} \frac{\phi(d_2)}{S^2\sigma\sqrt{T}} \tag{2.6}$$

based on the BS model. Gamma is the greatest at approximately at-the-money because delta as the probability proxy is expected to change significantly. However, gamma diminishes as the option becomes either more in-the-money or out-of-the-money. In other words, gamma is also important because it is the convexity of value. Gamma-neutral objective is dispensable for a delta-neutral portfolio.

Vega measures the rate of change of option value with respect to changes in its volatility $\nu = \frac{\partial f}{\partial \sigma}$, and

$$\nu_{BS} = Se^{-qT}\phi(d_1)\sqrt{T} = Ke^{-rT}\phi(d_2)\sqrt{T} \tag{2.7}$$

based on the BS model. Vega is also an important Greek to monitor especially in volatile markets.

Theta measures the rate of change of option value with respect to changes in its time to maturity $\theta = -\frac{\partial f}{\partial \tau}$, and

$$\begin{aligned} \theta_{BS_{\text{call}}} &= -e^{-qT} \frac{S\phi(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}\Phi(d_2) + qSe^{-qT}\Phi(d_1) \\ \theta_{BS_{\text{put}}} &= -e^{-qT} \frac{S\phi(d_1)\sigma}{2\sqrt{T}} + rKe^{-rT}\Phi(-d_2) - qSe^{-qT}\Phi(-d_1) \end{aligned} \tag{2.8}$$

respectively for call and put options based on the BS model. The option value consists of its intrinsic value and time value, and theta represents its time decay.

Rho measures the rate of change of option value with respect to changes in the interest rate $\rho = \frac{\partial f}{\partial r}$, and

$$\begin{aligned} \rho_{BS_{\text{call}}} &= KTe^{-rT}\Phi(d_2) \\ \rho_{BS_{\text{put}}} &= -KTe^{-rT}\Phi(-d_2) \end{aligned} \tag{2.9}$$

respectively for call and put options based on the BS model. Rho is only significant in extreme market conditions so it is the least popular first-order Greek.

2.4 Criticism of the BS framework

Obviously, the model simplifying assumptions are also its disadvantages, and the primary one is that the BSM framework assumes that the underlying asset price has constant volatility. In the real world, it is not.

There are a number of research papers attempting to address this problem in applying the BSM model to practice, and the implied volatility surface is one of them. The Hull and White (2017) also applied this pattern with an an empirical approach. The next chapter will discuss on previous research papers on the implied volatility surface.

Chapter 3

The Implied Volatility Surface

In the BS model, the volatility is an important parameter but it cannot be observed directly from markets as the others. Traders have to estimate the volatility from market data. They commonly view the volatility as both expected volatility and market sentiment for the underlying asset.

However, due to the impractical assumption of the BS model, many researchers have suggested concepts and models to circumvent this disadvantage, including the implied volatility surface.

3.1 The implied volatility

The only unobservable variable in the BS model is the volatility σ. It is also the most important parameter and is where the main flaw of the model lies. The variable can be numerically found by equating the observed option market prices (C_t, P_t) to their BS prices, which is called as the implied volatility. C_t and P_t are market prices for a call option and a put option respectively

$$\tilde{\sigma}^{IV} : BS_t(S_t,\ \tilde{\sigma}^{IV},\ cp\ flag,\ K,\ T,\ r,\ q) = C_t \mathbb{1}_{cp\ flag=1} + P_t \mathbb{1}_{cp\ flag=0}. \tag{3.1}$$

In addition, based on Equation 2.7, the BS vega of at time t can be expressed as

$$\nu_t^{BS} = \frac{\partial BS_t}{\partial \sigma} = S_t e^{-qT}\phi(d_1)\sqrt{T} = Ke^{-rT}\phi(d_2)\sqrt{T} > 0 \tag{3.2}$$

where $\phi(x)$ is the probability density function for a variable with a standard normal distribution. The BS vega is strictly positive so the BS price is monotone in σ. Hence, the $\tilde{\sigma}^{IV}$ solution to Equation 3.1 is unique.

3.2 The Volatility Smile

In reality, the implied volatility is not constant as it is assumed in the BS model, but it is in curvature patterns across different strikes K and terms T.

Definition 3.1 (IVS in absolute variables) *The mapping*

$$\tilde{\sigma}_t^{IV} : (K, T) \longmapsto \tilde{\sigma}_t^{IV}(K, T) \tag{3.3}$$

is called the implied volatility surface (IVS) in absolute variables.

Definition 3.2 (Volatility Smile) *For a given time to maturity T, Equation 3.3 is named the volatility smile.*

Specifically, many researchers have proved that there is a negative relationship between an equity price and its implied volatility. There are two main explanations for the phenomenon. The first reason is leverage, which argues that leverage decreases (increases) when the equity price increases (decreases), eventually, volatility decreases (increases) (Black 1976; Christie 1982; Bollerslev et al. 2006; Hens and Steude 2009; Hasanhodzic and Lo 2011). The other method, called as volatility feedback effect, explains that increasing (decreasing) volatility makes the required rate of return higher (lower), which discounts the equity price to a lower (higher) price level (French et al. 1987; Campbell and Hentschel 1992; Bekaert and Wu 2000). Empirical studies seem to, more or less, support the volatility feedback effect (Hull and White 2017).

Regardless of mechanism behind the volatility-equity price relationship, a large number of research papers has been motivated to relax the assumptions of the original BS model, resulting in alternative models for more effective options pricing and hedging. Stochastic volatility (SV) is one of the important alternative model classes, in which volatility follows a stochastic process. Bollerslev et al. (2006) pointed out that several popular continuous-time SV models effectively described the observed volatility asymmetries. Classical SV models were developed by Hull and White (1987), Hull and White (1988), Heston (1993), and Hagan et al. (2002).

3.3 Smile-adjusted delta hedging

Delta hedging is the most important practice in risk management of option portfolios as it mitigates the risk against price movements in the underlying asset.

Hedging volatility $\hat{\sigma}_h$ is the volatility that equates the BS delta with the delta of the true but unknown pricing model.

$$\hat{\sigma}_h\colon \qquad \frac{\partial f_t(\hat{\sigma}_h)}{\partial S_t} - \frac{\partial \tilde{BS}_t}{\partial S_t} = 0 \tag{3.4}$$

where f is the option price determined by its true underlying model with $\hat{\sigma}_h$ as volatility.

Practitioners usually plug $\tilde{\sigma}_t^{IV}$ into the BS delta $\delta_t^{BS} = \partial BS_t/\partial S_t$ for option positions and this is called the BS hedging method. However, this method does not always provide an effective hedging strategy because the implied volatility $\tilde{\sigma}_t^{IV}$ is sometimes not equal to the hedging volatility $\hat{\sigma}_h$. For example, given the Hull and White SV model (1987) as the true underlying price process, Renault and Touzi (1996) proved that the BS delta hedging method under-hedged the portfolio for in-the-money (ITM) options and over-hedged that for out-of-the-money (OTM) options. It was effective only for at-the-money (ATM) options.

Nonetheless, the genuinely true underlying model such that $\{\hat{\sigma}_h\}_t$ perfectly hedges a portfolio of options rarely exists due to complexity of our real world. In particular, in the case of SV, the market is incomplete and there is no perfect hedge (Alexander, Rubinov, et al. 2012). Therefore, we need a sub-optimal measure for hedging performance.

Definition 3.3 *Minimum variance (MV) delta δ_{MV} gives a position in the underlying asset that minimizes the instantaneous variance of hedging error*

$$\delta_{MV} \coloneqq \arg\min_{\delta_{MV}} \mathbb{V}[\Delta f - \delta_{MV}\Delta S], \tag{3.5}$$

where Δf and ΔS are price changes in an option and its underlying respectively.

Since the early implementation of the above hedging analysis method by Ederington (1979), many researchers have succeeded to convert the BS delta to MV deltas to improve hedging performance with assumptions of the classical SV models. Bakshi et al. (1997) empirically compared the MV delta hedging performance among the Heston SV model (1993), a stochastic-volatility random-jump (SVJ) model, a stochastic-volatility and stochastic-interest-rate (SVSI) model, and the BS model, using data on call options on the S&P 500 index between June 1988 and May 1991. They showed that the SV model was the best model while the BS model was between the worst two. Later, Bakshi et al. (2000) also showed that the SV outperformance was even more significant for long-term options on the S&P 500 index between September 1993 and August 1995. With the Heston SV model (1993) and the SABR model (Hagan et al. 2002) as the true underlying models, Alexander, Kaeck, et al. (2009) and Poulsen et al. (2009) came to the same conclusion using call and put options on the S&P 500 index, the Eurostoxx index, and USD/EUR exchange rate between 2004 and 2008.

There were also other similar studies but they were not based on any SV model. These authors assumed that the MV delta was equal to the BS delta plus the BS vega times the partial derivative of the expected implied volatility with respect to either the strike price or the underlying price. On the one hand, with the local volatility model (Derman et al. 1996; Dupire et al. 1994), Crépey (2004) presented hedging improvement for call and put options on the FTSE 100 during 1999-2000 and the DAX during 2001, arguing the partial derivative was equal to the slope of the volatility smile $\partial\mathbb{E}(\sigma^{IV})/\partial K$. With the same approach, Vähämaa (2004) extended the empirical evidence that the adjustment was effective in hedging for the FTSE 100 index during 2001.

Although the relationship in Equation 3.3 indicating the IVS depends on the underlying price S_t is not obvious, Lee (2005) proved that $\partial\sigma^{IV}/\partial S_t = -(K/S_t)(\partial\sigma^{IV}/\partial K)$ in a SV framework. Derman (1999) stated hypotheses that the volatility dynamics varies with different market regimes describing characteristics of returns and volatility. The regimes are related to different time periods. Then, Alexander, Rubinov, et al. (2012) carried out an extensive investigation with eight different models for the partial derivative with respect to the underlying price S_t, including the different market-regime models using daily options data on the FTSE 100 index between 1993 and 2009. One of the three best models had its MV delta called "regime-dependent smile-adjusted delta", and its adjustment was computed from the current smile and a short history of ATM volatility.

Definition 3.4 *An alternative approach for approximating the MV delta is*

$$\delta_{MV_t} = \delta_{BS_t} + \nu_{BS_t} \frac{\partial \mathbb{E}(\tilde{\sigma}^{IV})}{\partial S} \tag{3.6}$$

by explicitly assuming that the smile depends on the underlying asset price S_t. This is exactly true when both the underlying price S_t and the implied volatility are diffusion processes, but it is just an approximation in other cases (Hull and White 2017).

Notably and similarly, Hull and White (2017) proposed an approach empirically estimating the partial derivative with a quadratic function of the BS delta δ^{BS}. They also broadly tested the framework on call and put options on the S&P 500, S&P 100, the Down Jones Industrial Average of 30 stocks (DJIA) indices, the individual stocks underlying the DJIA, a gold ETF, a silver ETF, an oil ETF, the Barclays U.S. 20+ year Treasury Bond Index TLT) and the Barclays U.S. 7-10 year Treasury Bond Index (IEF). The period covered was from 2004 to 2015, except the commodity ETFs from 2008. The hedging gains of the MV delta over the BS delta were greater than the SABR model (Hagan et al. 2002), and a local volatility model (Derman et al. 1996; Coleman et al. 2003).

However, the results were still limited by noises in the assumed relationship between changes in the underlying price and changes in its volatility. Hence, the next chapter will explain the gradient boosting method that this study uses to improve the Hull and White empirical model in hedging performance.

Chapter 4

Supervised learning

Statistical learning is a process attempting to solve problems based on data. Given inputs X, it will return outputs Y. Supervised learning adapts and increases its knowledge based on received feedback, which is called model training. In unsupervised learning, feedback is not available.

A statistical model for the joint distribution of X and Y addresses uncertainty between input and output variables. Trevor et al. (2009) introduced supervised learning in a context of an additive error model $Y = f(X) + \varepsilon$ by assuming ε is independent of X and $\mathbb{E}[\varepsilon] = 0$. They also classified supervised learning models into two main categories according to the structural assumptions of the models. The first category of local methods contains unstable models (high variance, low bias) that suffer form the curse of dimensionality. The second category consists of more structured regression models that are more stable (low variance, high bias).

4.1 Regression trees

Decision trees is a local method with high variance and low bias as classified by Trevor et al. (2009) earlier. Decision tree model consists of classification trees and regression trees.

Definition 4.1 (Decision tree) *The classification and regression tree (CART) is*

$$h_\Theta(x) = \sum_{j=1}^{M} c_j \mathbb{1}_{x \in S_j} \tag{4.1}$$

where $\Theta = \{S_j, c_j\}_{j=1}^{M}$ follows a target criterion (Breiman et al. 1984).

Definition 4.2 (Regression tree) *In the regression framework, both the response variable Y and the predictor variable X are continuous. S_j is of the form $S_j(u, v) = \{X \in \mathbb{R}^p | X_u \leq v\}$. X_u is called the split variable, and v is the cut value (Breiman et al. 1984).*

Each component of the predictor variable X_u is checked for a best cut value v_u such that the resulting two groups are homogeneous with respect to the response variable $y_i \in Y$. The split that yields the smallest variance within a group is chosen.

This procedure is repeated, leading to a sequence of binary splits that forms a maximal regression tree. The tree is then pruned by a cross-validation scheme to prevent it from over-fitting.

Given we adopt the criterion as minimization of the sum of squared errors $\sum(y_i - h(x_i))^2$, $\hat{S}_j(u, v)$ is determined in each step of the iteration such that

$$\min_{u,v} \left[\min_{c_{j_1}} \sum_{i:\, x_i \in S_j(u,v)} (y_i - c_{j_1})^2 + \min_{c_{j2}} \sum_{i:\, x_i \notin S_j(u,v)} (y_i - c_{j_2})^2 \right]. \tag{4.2}$$

In addition, it is noted that c_{j_1} and c_{j_2} are equal to simple averages of $y_{i_{\{x_i \in S_j\}}}$ and $y_{i_{\{x_i \notin S_j\}}}$ respectively as following

$$\hat{c}_{j_1} = \frac{1}{\sum_{i=1}^{N} \mathbb{1}_{x_i \in S_j}} \sum_{i=1}^{N} y_i \mathbb{1}_{x_i \in S_j} \tag{4.3}$$

and

$$\hat{c}_{j_2} = \frac{1}{\sum_{i=1}^{N} \mathbb{1}_{x_i \notin S_j}} \sum_{i=1}^{N} y_i \mathbb{1}_{x_i \notin S_j}. \tag{4.4}$$

4.2 Boosting and addictive trees

Ensemble learning firstly develops a collection of base learners from the training data, and then combines them to form a composite predictor. Ensemble learning is classified into bagging and boosting. On the one hand, many independent learners are trained in the bagging method, and the composite predictor is some average of their predictions. On the other hand, learners are trained sequentially in boosting so that subsequent learners learn from mistakes of the previous ones. Hence, boosting is less computationally expensive for the loss function to converge because the mistakes are focused in learning. (Trevor et al. 2009).

Among boosting methods, gradient boosting is very popular. Decision trees are typically chosen to be the base learners because of its ability to handle data of mixed types and to model complex functions. In this study, we use gradient boosting regression trees (GBRTs) method to estimate the further adjustments to the MV hedging residuals and improve hedging performance.

Definition 4.3 (GBRT) *GBRTs consider additive models of the following form:*

$$F(x) = \sum_{m=1}^{M} \gamma_m h_m(x) \tag{4.5}$$

where $h_m(x)$ is a regression tree as the base learner and γ_m is a scalar.

At each iteration m, a regression tree $h_m(x)$ is added such that the loss function L is minimized

$$h_m(x) = \arg\min_h \sum_h^n L\left(y_i, F_{m-1}(x_i) + h_m(x_i)\right). \tag{4.6}$$

Based on a greedy approximation method proposed by Friedman (2001), h_m is estimated to be the steepest descent equal to gradient of $L(F_{m-1})$ with respect to F_{m-1}

$$h_m = -\left[\frac{\partial L(y_i, F_{m-1}(x_i))}{\partial F_{m-1}(x_i)}\right]. \tag{4.7}$$

Provided the least squares of error is the loss function, the gradient is proportional to the residuals $y_i - F(x_i)$.

Chapter 5

The model

In this chapter, we discuss the idea of the MV model and provide the related descriptive analysis to options on the S&P 500 index between 2015 and 2017. A classical boosting algorithm is introduced to determine an adjustment to the MV delta to further improve hedging performance.

5.1 Data

We use data from OptionMetrics to investigate options on the S&P 500 index. The period covered by the data is from January 2, 2015 to December 29, 2017. Option price f is option last price if volume is positive; otherwise, option price is equal to mid price. Only observations with valid bid price and offer price are retained. Those having zero open interest are also removed.

Option series are classified by their unique strikes and expiration dates. Those having only one observation are removed from the dataset. Options with remaining lives less than 14 days are also removed. Only those with absolute values of their BS deltas between 0.05 and 0.95 are retained.

Eventually, there are about 1.3 million observations for nearly 17,000 call and 22,000 put option series, as presented in Table 5.1. Options with maturities less than 91 days constitute 60 percent of the dataset. Table 5.2 demonstrates that there are about 35 observations per option series on average for both call and put options. The largest number of observations is greater than 600.

Table 5.3 shows that trading volume of put options was higher than of call options by 50 percent . ATM and OTM options were more significantly actively traded than ITM ones. Deep OTM put options were popular in all maturities while ATM and deep ITM call options expired within 91 days were the most traded.

Number of observations

	Maturity in days						
	< 91		92-365		> 365		
Moneyness	Call	Put	Call	Put	Call	Put	Total
0.1	53,440	154,133	21,026	65,583	11,409	33,034	338,625
0.2	30,598	73,322	14,119	33,651	7,891	15,854	175,435
0.3	25,993	47,446	12,190	22,360	6,507	10,179	124,675
0.4	25,557	36,319	11,219	16,472	5,796	8,972	104,335
0.5	27,126	29,987	11,625	12,958	6,140	7,772	95,608
0.6	29,831	25,416	12,824	10,154	7,258	5,793	91,276
0.7	33,257	24,412	13,905	9,466	9,008	4,408	94,456
0.8	40,972	37,150	17,296	13,376	9,862	4,653	123,309
0.9	79,249	36,456	22,093	13,837	10,646	3,683	165,964
Total	346,023	464,641	136,297	197,857	74,517	94,348	1,313,683

Table 5.1: Number of observations of the S&P 500 index options from January 2, 2015 to December 29, 2017 (754 days in total). Moneyness is measured by $|\delta_{BS}|$ rounding to the nearest ten. Maturity is measured in calendar days. From small to large $|\delta_{BS}|$ values, the moneyness column represents the following moneyness categories: deep out-of-the-money (DOTM), out-of-the-money (OTM), at-the-money (ATM), in-the-money (ITM), and deep in-the-money (DITM).

Number of observations by option series

	mean	std	min	25%	50%	75%	max
call	34	51	2	9	22	40	670
put	36	51	2	10	24	43	644

Table 5.2: Number of observations of the S&P 500 index options by option series from January 2, 2015 to December 29, 2017 (754 days in total). Moneyness is measured by $|\delta_{BS}|$ rounding to the nearest ten. Maturity is measured in calendar days. From small to large $|\delta_{BS}|$ values, the moneyness column represents the following moneyness categories: deep out-of-the-money (DOTM), out-of-the-money (OTM), at-the-money (ATM), in-the-money (ITM), and deep in-the-money (DITM).

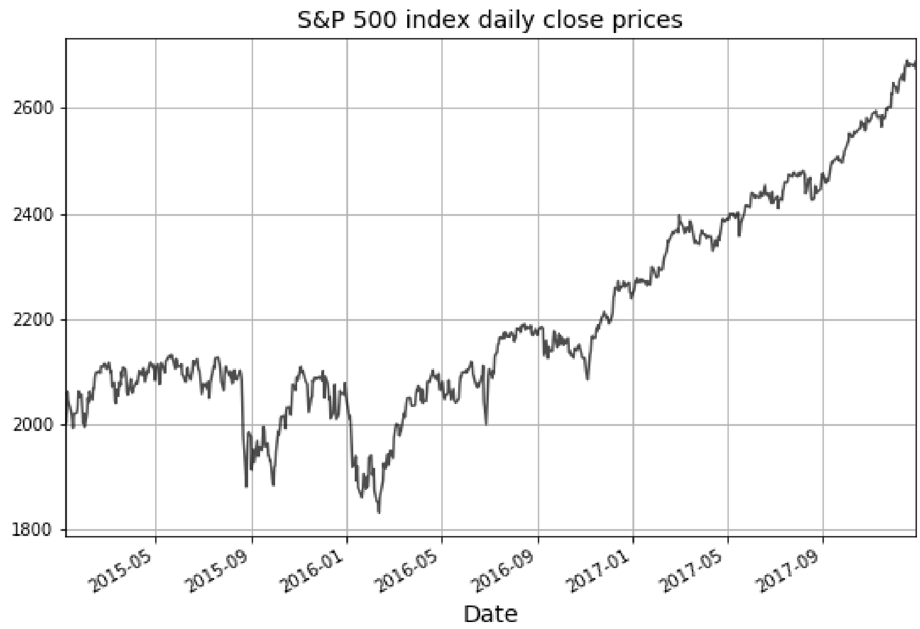
Trading volume

	Maturity in days						
	< 91		92 - 365		> 365		
Moneyness	Call	Put	Call	Put	Call	Put	Total
0.1	22.2	50.5	3.1	9.4	0.3	0.9	86.4
0.2	16.3	30.7	3.1	7.7	0.2	1.0	58.9
0.3	12.9	19.5	2.4	4.9	0.3	0.6	40.6
0.4	11.4	20.8	2.0	4.7	0.2	0.7	39.7
0.5	23.3	22.9	4.0	3.7	0.4	0.2	54.5
0.6	12.3	3.5	1.8	0.3	0.3	0.0	18.3
0.7	2.6	1.2	0.4	0.1	0.1	0.0	4.5
0.8	1.2	0.6	0.2	0.1	0.1	0.0	2.1
0.9	0.9	0.2	0.3	0.1	0.1	0.0	1.5
All	103.0	149.8	17.4	30.9	2.0	3.5	306.5

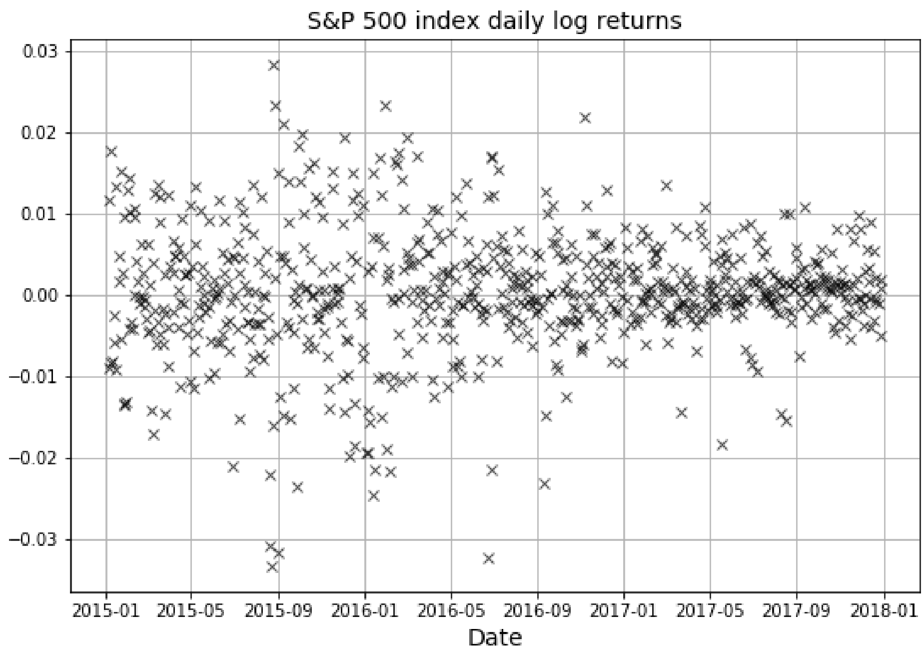
Table 5.3: Trading volume of the S&P 500 index options from January 2, 2015 to December 29, 2017 (754 days in total). Moneyness is measured by $|\delta_{BS}|$ rounding to the nearest ten. Maturity is measured in calendar days. From small to large $|\delta_{BS}|$ values, the moneyness column represents the following moneyness categories: deep out-of-the-money (DOTM), out-of-the-money (OTM), at-the-money (ATM), in-the-money (ITM), and deep in-the-money (DITM).

5.2 Descriptive analysis of S&P 500 options

From now, the subscript t is left out for shorter abbreviations. The MV delta is investigated for options on the S&P 500 index in order to testify the functional form suggested by Hull and White (2017) between 2015 and 2017. We start with



(a) S&P 500 index daily close prices



(b) S&P 500 index daily log returns

Figure 5.1: Daily close price and log returns of S&P 500 index

Equation 3.5 applied to daily price changes:

$$\Delta f = \delta_{MV} \Delta S + \varepsilon \tag{5.1}$$

where ε is an error term. The MV delta δ_{MV} is supposed to minimize variance of error term ε, which is proportional to its sum of squares. Thus, we can estimate descriptively δ_{MV} using regression. Hull and White (2017) attempted other variations on the model. For example, they replaced Δf with $\Delta f - \theta_{BS} \Delta t$, where $\theta_{BS} = \partial f / \partial t$ is the BS theta and Δt is one trading day. In addition, they also used non-normalized data, and included an intercept. Nonetheless, they found no material differences in their results.

The same settings are used in this study. We estimate δ_{MV} in Equation 5.1 for options with different moneyness and time to maturity. Moneyness is measured by δ_{BS} and it is grouped into rounding δ_{BS} to the nearest ten, particularly, $[0.1, 0.2, ..., 0.9]$ for call options and $[-0.1, -0.2, ..., -0.9]$ for put options. Option maturity buckets are 14-30, 31-60, 61-91, 92-122, 123-182, 183-365, and more than 365 days.

Remark 5.1 Figure 5.2 empirically shows that $(\delta_{MV} - \delta_{BS})$ values are all negative for both call and put options because ν_{BS} is strictly positive (Equation 3.2). This indicates that the BS delta over-hedges call options and under-hedges put options on the S&P 500 index compared to the MV delta. This result is consistent with the literature on the negative relationship between the underlying price and its volatility following the volatility smile.

Remark 5.2 From 2015 to 2017, $(\delta_{MV} - \delta_{BS}) S\sqrt{T} / \nu_{BS}$ is roughly quadratic in δ_{BS} separately for call and put options on the S&P 500 index (Figure 5.2), while it is insignificantly dependent on option maturity (Figure 5.3). Then, we have

$$\delta_{MV} \approx \delta_{BS} + \frac{\nu_{BS}}{S\sqrt{T}}(a + b\delta_{BS} + c\delta_{BS}^2) \tag{5.2}$$

where a, b, and c are constants. From Equation 3.6, we can show that

$$\mathbb{E}(\Delta\sigma_{imp}) \approx \left(\frac{a + b\delta_{BS} + c\delta_{BS}^2}{\sqrt{T}}\right)\frac{\Delta S}{S} \tag{5.3}$$

Note 5.1 Hull and White (2017) earlier proved Equation 5.2 with the assumption of scale invariance.

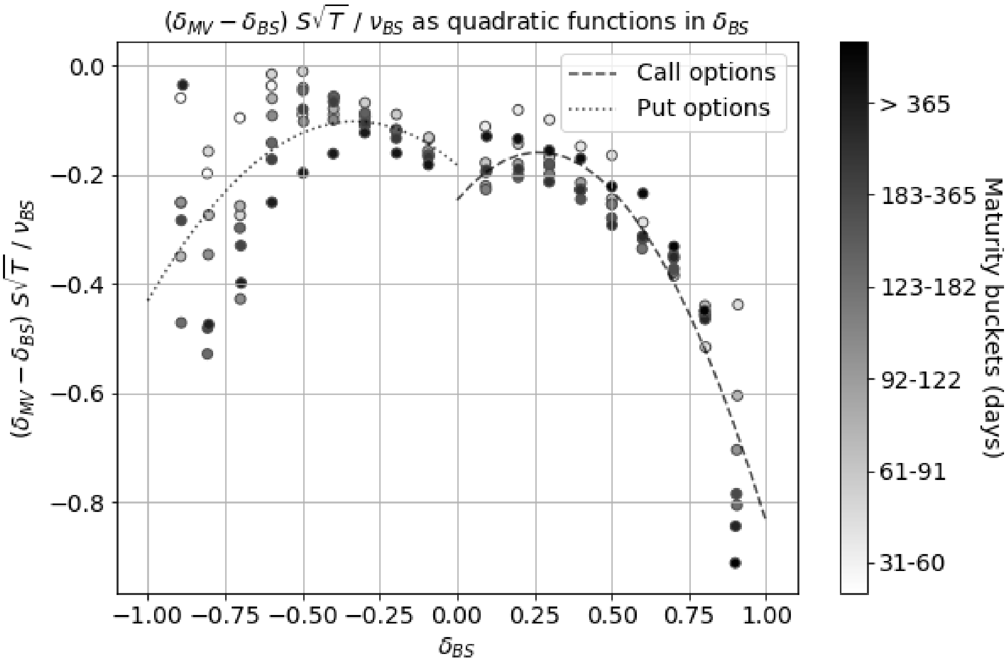


Figure 5.2: $(\delta_{MV} - \delta_{BS})S\sqrt{T}/\nu_{BS}$ was roughly quadratic in δ_{BS} buckets separately for call and put options on the S&P 500 index with different moneyness and time to maturity between 2015 and 2017. δ_{MV} is separately estimated by $\Delta f = \delta_{MV}\Delta S + \varepsilon$ using regression with the mean-squared error (MSE) method. Options with remaining lives less than 14 days, and absolute values of the BS delta less than 0.05 or greater than 0.95 are removed.

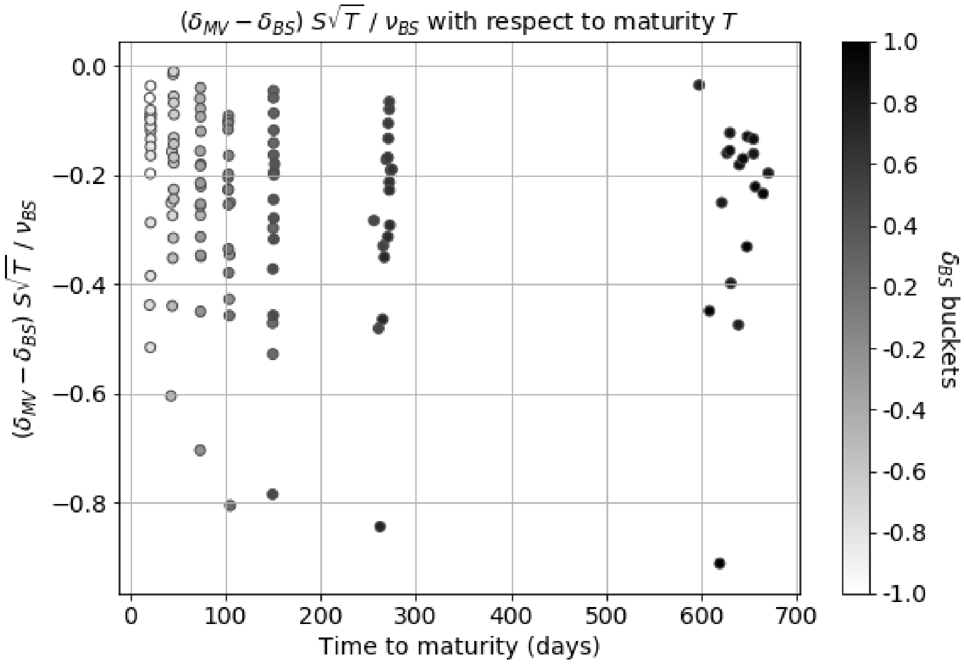


Figure 5.3: $(\delta_{MV} - \delta_{BS})S\sqrt{T}/\nu_{BS}$ was independent of option maturity T for call and put options on the S&P 500 index with different moneyness and time to maturity between 2015 and 2017. δ_{MV} is separately estimated by $\Delta f = \delta_{MV}\Delta S + \varepsilon$ using regression with the mean-squared error (MSE) method. Options with remaining lives less than 14 days, and absolute values of the BS delta less than 0.05 or greater than 0.95 are removed.

5.3 The model

5.3.1 The starting model

Equation 5.3 provides that, given a particular underlying price change at particular moneyness and maturity, the expected change in the implied volatility is a quadratic function of the moneyness measured by δ_{BS}. The unknown element of our model is the quadratic function of moneyness. We estimate the model parameters a, b and c using a regression model based on Equation 3.6 and 5.2.

$$\Delta f - \delta_{BS}\Delta S = \frac{\nu_{BS}}{\sqrt{T}}\frac{\Delta S}{S}(a + b\delta_{BS} + c\delta_{BS}^2) + \varepsilon \tag{5.4}$$

We use a moving window where the parameters are estimated by separately for all call and put options over rolling and overlapping 24-month periods. Then, we use the parameters to determine MV hedges during the following month. We also tested shorter regression periods than 24 months but their resulting parameters were not stable. In addition, we should not choose any longer period because our whole data period covers only 36 months. Therefore, the 24-month period is reasonable in this context.

Remark 5.3 The estimated coefficients $\hat{a}$, $\hat{b}$, and $\hat{c}$ are shown in Figure 6.2. The parameters are highly stable as they change gradually through time. The last four months in 2017 witness more significant monthly changes in $\hat{b}$ and $\hat{c}$. This implies that there was a structural change in the relationship between δ_{BS} and ε_{BS} from September 2017. Hence, predicting power of the MV model is likely to deteriorate towards the year end.

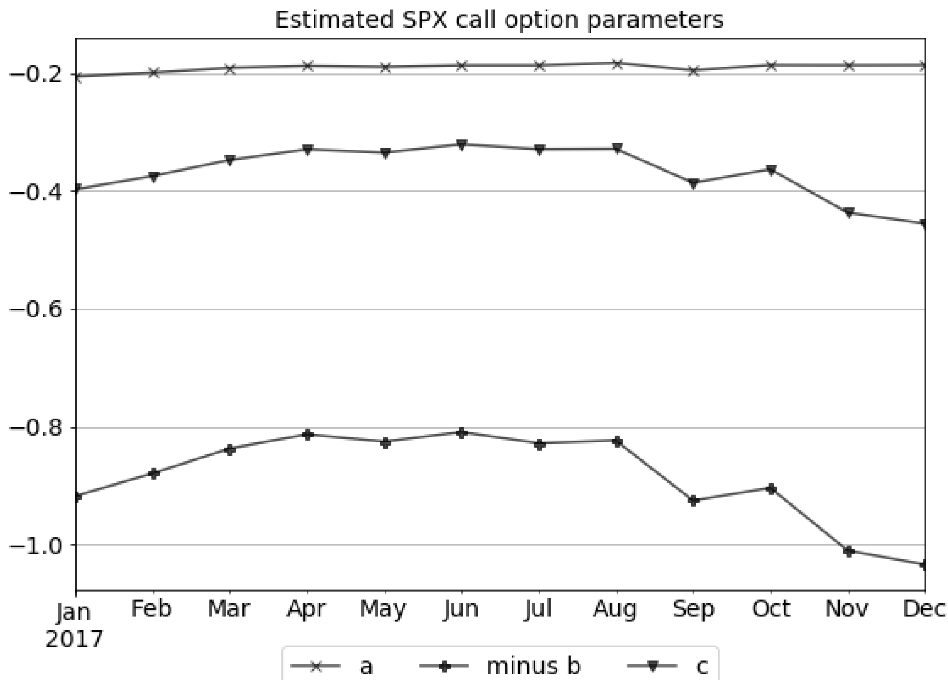
5.3.2 Improve upon the starting model

Definition 5.1 *Hedging error ε_{BS} based on the BS delta is*

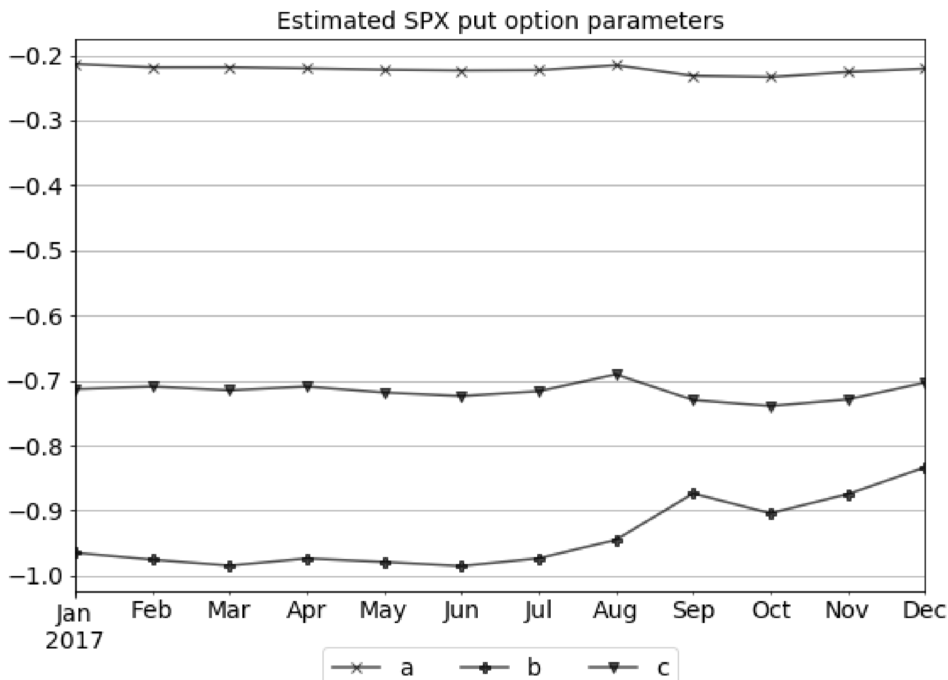
$$\varepsilon_{BS} = \Delta f - \delta_{BS}\Delta S. \tag{5.5}$$

Definition 5.2 *Hedging error ε_{MV} based on the MV delta calculated with Equation 5.2 is*

$$\varepsilon_{MV} = \varepsilon_{BS} - \frac{\nu_{BS}}{\sqrt{T}}\frac{\Delta S}{S}(\hat{a} + \hat{b}\delta_{BS} + \hat{c}\delta_{BS}^2) \tag{5.6}$$



(a) Estimated call parameters



(b) Estimated put parameters

Figure 5.4: Estimated quadratic coefficient parameters using regression with Equation 5.4 for call and put options on the S&P 500 index in 2017. Rolling and overlapping 24-month periods are used to estimate.

where $\hat{a}$, $\hat{b}$, and $\hat{c}$ are the estimated coefficients from Equation 5.4 using regression.

Note 5.2 Intuitively, we firstly have $\varepsilon_{MV} = \Delta f - \delta_{MV}\Delta S$. By substituting δ_{MV} from Equation 5.2, Equation 5.6 can be obtained as the following

$$\begin{aligned}\varepsilon_{MV} &= \Delta f - \left[\delta_{BS} + \frac{\nu_{BS}}{S\sqrt{T}}(\hat{a} + \hat{b}\delta_{BS} + \hat{c}\delta_{BS}^2)\right]\Delta S \\ &= \Delta f - \delta_{BS}\Delta S - \frac{\nu_{BS}}{\sqrt{T}}\frac{\Delta S}{S}(\hat{a} + \hat{b}\delta_{BS} + \hat{c}\delta_{BS}^2) \\ &= \varepsilon_{BS} - \frac{\nu_{BS}}{\sqrt{T}}\frac{\Delta S}{S}(\hat{a} + \hat{b}\delta_{BS} + \hat{c}\delta_{BS}^2).\end{aligned}$$

Remark 5.4 Hence, the MV model actually determines adjustments into the BS residuals to result in MV residuals with minimum variance. Similarly, this research proposes a method to further adjust those MV residuals to improve hedging performance with the boosting.

Definition 5.3 *Let F_0 denote a function that further adjusts the MV residuals ε_{MV} in Equation 5.6 to reduce variance of hedging residuals. $cp\ flag$ is also one of its independent variables because parameters are approximated separately for call and put options.*

$$\varepsilon_{MVB} = \varepsilon_{MV} - F_0(\cdot, cp\ flag) \tag{5.7}$$

where ε_{MVB} is hedging errors improved by the model F_0 with $\mathbb{E}[\varepsilon_{MVB}] = 0$ and $\mathbb{E}[\varepsilon_{MVB}^2] < \infty$.

Note 5.3 An indicator function incorporates the $cp\ flag$ variable into F_0 as below

$$F_0(\cdot, cp\ flag) = F_0(\cdot)\mathbb{1}_{cp\ flag=call} + F_0(\cdot)\mathbb{1}_{cp\ flag=put}. \tag{5.8}$$

We apply the classical boosting algorithm on the residuals ε_{MV} to further reduce its variance, thus improve hedging performance of the MV model. Beforehand, we need to define a predictor space for the algorithm.

Definition 5.4 *Due to its property of feature selection, we include a variety of observed basic variables into the predictor space and let the algorithm allocate optimal weights onto them. The included variables are the BS delta δ_{BS}, the implied volatility change $\Delta\sigma_{imp}$, underlying price change ΔS, time to maturity in annual term T, and $cp\ flag$.*

$$\mathrm{x} \coloneqq (\delta_{BS}, \Delta\sigma_{imp}, \Delta S, T, cp\ flag) \tag{5.9}$$

Remark 5.5 Changes in underlying price ΔS and option price Δf cannot be both included in the predictor space because hedging residuals $\varepsilon = \Delta f - \delta \cdot \Delta S$ are just directly related to these observable variables. Hence, only underlying price change ΔS is included to prevent the algorithm from over-fitting.

From Equation 5.7, we propose a general model with an unspecified function f on the predictor space x

$$\varepsilon_{MVB} = \varepsilon_{MV} - f(\mathrm{x}). \tag{5.10}$$

Our framework is expected to minimize variance of hedging errors through a linear additive expansion of the form

$$f(\mathrm{x}) = F_0(\mathrm{x}) + \sum_{j=1}^{M} B_j(\mathrm{x}) \tag{5.11}$$

where each B_j denotes a base learner. We choose regression trees as the base learner because of their advantages (Section 4.2).

Remark 5.6 δ_{MVB} can be computed from $f(\mathrm{x})$ as following

$$\delta_{MVB} = \delta_{MV} + \frac{f(\mathrm{x})}{\Delta S} \tag{5.12}$$

Definition 5.5 *The proposed methodology relies on a quadratic loss function which is proportional to variance of the MVB hedging errors* ε_{MVB}

$$L\left(\varepsilon_{MV}, f(\mathrm{x})\right) = \sum_{t=1}^{N}\sum_{i=1}^{L_t}(\varepsilon_{MV_{ti}} - f(\mathrm{x}_{ti}))^2 \tag{5.13}$$

where L_t *is the number of observations varying per day* $t \in \{1, ..., N\}$.

Algorithm: Tree-boosting for the MV hedging errors

In-sample period is the 24 months used to estimate the parameters consisting of N trading days. The original MV model by Hull and White (2017) is fitted into the in-sample data to estimate the MV residuals $\hat{\varepsilon}_{MV}$.

1. Remove $\hat{\varepsilon}_{MV}$ and x outliers whose values are outside their $[Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR]$ ranges. Q_1 and Q_3 are their 25-percentile and 75-percentile values respectively. $IQR = Q_3 - Q_1$ are their interquantile ranges.

2. Randomly split the in-sample period into a learning sample period LS with $[0.8 \cdot N]$ observations and a validation sample period VS with $N - [0.8 \cdot N]$ observations.

3. For $j = 0, ..., M$:

   (a) Compute the residuals for all options in the learning sample

   $$\text{residuals}_{LS} = \hat{\varepsilon}_{MV_{LS}} - \hat{F}_j(\mathrm{x}_{LS}).$$

(b) Fit regression tree$^{\mathrm{c}}$ to $\{\text{residuals}_{LS} | cp\ flag_{LS} = 1\}$ for all call options in the learning sample $\rightsquigarrow \widehat{\text{tree}^{c}}(\mathrm{x})$.

(c) Fit regression tree$^{\mathrm{p}}$ to $\{\text{residuals}_{LS} | cp\ flag_{LS} = 0\}$ for all put options in the learning sample $\rightsquigarrow \widehat{\text{tree}^{p}}(\mathrm{x})$.

(d) Base learner j is then given by

$$\hat{B}_j(\mathrm{x}) = \widehat{\text{tree}^{c}}(\mathrm{x}) \mathbb{1}_{cp\ flag=1} + \widehat{\text{tree}^{p}}(\mathrm{x}) \mathbb{1}_{cp\ flag=0}.$$

(e) Update

$$\hat{F}_{j+1}(\mathrm{x}) = \hat{F}_j(\mathrm{x}) + \eta \hat{B}_j(\mathrm{x})$$

with small shrinkage factor $0 < \eta \leq 1$.

4. Choose j such that $L\left(\hat{\varepsilon}_{MV}, \hat{F}_j(\mathrm{x})\right)$ is minimal on the validation sample. $\hat{M}$ is the optimal stopping value such that

$$\hat{M} = \arg\min_j L\left(\hat{\varepsilon}_{MV_{VS}}, \hat{F}_j(\mathrm{x}_{VS})\right). \tag{5.14}$$

5. Repeat Steps 3 and 4 for the whole in-sample data, including both the learning and validation samples with $\hat{M}$ iterations. An estimate for the unspecified function f in the general model of Equation 5.10 is then given by

$$\hat{f}(\mathrm{x}) = \hat{F}_{\hat{M}}(\mathrm{x}) = \eta \sum_{j=1}^{\hat{M}} \hat{B}_j(\mathrm{x}).$$

Remark 5.7 Outlier removal in the Step 1 above prevents the model from overfitting by reducing model variance. Otherwise, resulting relative importance of underlying price change would be unnecessarily greater to explain abnormal changes in hedging residuals of outlier observations.

Remark 5.8 (Default values) This study uses a learning rate $\eta = 0.1$ and $M = 500$ iterations.

Remark 5.9 The cross-validation in Step 3 requires us to evaluate $\hat{F}_j(\mathrm{x}_{ti})$ for all options in the validation sample to calculate $L(\hat{F}_j)$. Once the optimal stopping value $\hat{M}$ has been found, $F_{\hat{M}}$ needs to be estimated again, now using the whole sample and not only the first 80% of the data.

5.3.3 Boundaries

Boundaries are important in applications of statistical learning to mitigate model risks. This study employs two boundaries for the MVB delta.

The first boundary is $\delta_{MVB} \in [0, 1]$ for call options and $\delta_{MVB} \in [-1, 0]$ for put options.

The second boundary is used to limit daily changes in hedging position and save hedging costs in practice. The BS gamma as a typical proxy for the BS delta-neutral hedging cost is a reasonable factor to determine the limits. Practitioners often use BS percentage gamma $\%\gamma_{BS}$ instead of the BS gamma to think in percentage movements of underlying price rather than absolute changes (Equation 5.15). In this research, we restrict the MVB delta as an adjustment from the MV delta within $20 \cdot \%\gamma_{BS}$: $\delta_{MVB} \in [\delta_{MV} - 20 \cdot \%\gamma_{BS}, \delta_{MV} + 20 \cdot \%\gamma_{BS}]$.

Definition 5.6 *BS percentage gamma $\%\gamma_{BS}$ is a change in the BS delta δ_{BS} given a 1% change in underlying price S.*

$$\%\gamma_{BS} = \frac{\Delta\delta_{BS}}{0.01S} \tag{5.15}$$

The next chapter will present empirical results using the proposed approach for options on the S&P 500 index in 2017.

Chapter 6

Out-of-sample analysis of S&P 500 index options

This chapter details empirical results using the proposed model for options on the S&P 500 index in 2017. We also include our discussion and interpretation of the results.

6.1 OS performance measures

6.1.1 Hedging gain

Definition 6.1 *We denote Gain as hedging effectiveness equal to percentage reduction in sum of squared hedging errors. Gain of model A over the BS model is*

$$Gain_A = 1 - \frac{\sum \varepsilon_A^2}{\sum \varepsilon_{BS}^2}. \tag{6.1}$$

The Gain is calculated for different delta buckets, and then for all options. This results in nine bucketed Gains and one overall Gain for each test month separately for call and put options.

Remark 6.1 Using standard deviations rather than SSEs would produce a similar relative measure but the Gain would be numerically smaller.

Hull and White (2017) developed a procedure to calculate the t-statistic used to testify statistical significance of difference between the Gains for two different hedging models.

6.1.2 Statistical test of improvement

Definition 6.2 (Diff statistic) *Considering n observations, let x_i be the squared residual from hedging using model A, y_i be that using model B, and z_i be the squared residual from the BS model. The Gain from model A is*

$$Gain_A = 1 - \sum_{i=1}^{n} x_i \bigg/ \sum_{i=1}^{n} z_i$$

Similarly,

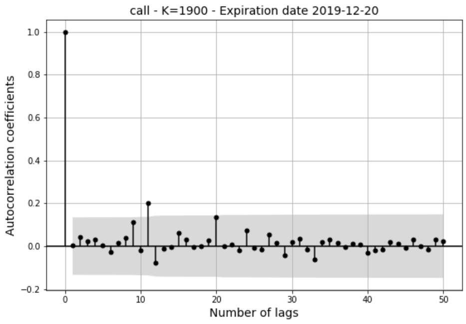
$$Gain_B = 1 - \sum_{i=1}^{n} y_i \bigg/ \sum_{i=1}^{n} z_i$$

Defining $Z = \sum_{i=1}^{n} z$, the difference in the Gains is

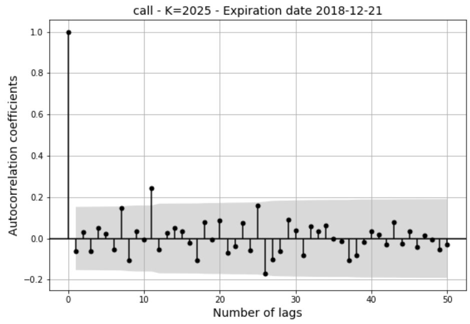
$$Diff = \frac{1}{Z} \sum_{i=1}^{n} (x_i - y_i) \tag{6.2}$$

Definition 6.3 (Sample mean) *Suppose the individual difference observations, $x_i - y_i$, are drawed from a distribution with mean μ and standard deviation σ. Firstly, the mean can be estimated from the observed Diff values.*

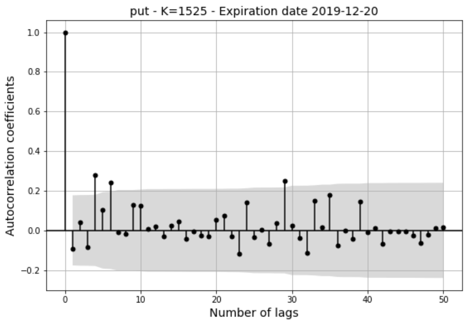
$$\hat{\mu} = \sum_{i=1}^{n} (x_i - y_i) \bigg/ n \tag{6.3}$$



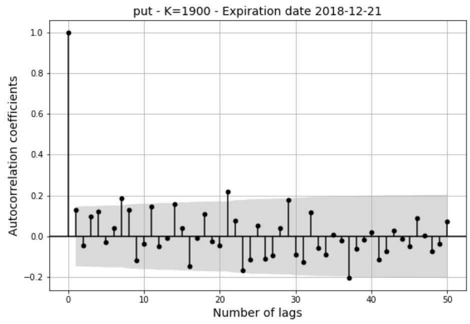
(a) Call K=1900 Expiration at 20 Dec 2019



(b) Call K=2025 Expiration at 21 Dec 2018



(c) Put K=1525 Expiration at 20 Dec 2019



(d) Put K=1900 Expiration at 21 Dec 2018

Figure 6.1: Autocorrelation of the Diff statistic for four sample option series with 95% confidence interval

Remark 6.2 Figure 6.1 shows that there are actually serial correlations in the Diff statistics for some option series on the S&P 500 index in 2017 with 95% confidence level. Hence, we should apply a Newey-West adjustment factor to the estimated variance.

Definition 6.4 (Newey and West (1986)) *The Newey-West adjustment factor for an option series, on which the i^{th} observation is based, is estimated as following*

$$\hat{w}_i(m) = 1 + 2\sum_{j-1}^{m-1}\left(\frac{m-j}{m}\right)\tilde{p}_j \tag{6.4}$$

where $\tilde{p}_j$ are $m-1$ correlation coefficients.

The Diff statistic is calculated for each option series with unique strike price and expiration date using Equation 6.2 with n equal to its number of observations, then the Newey-West adjustment factors for m lags between 1 and 50 are also estimated by Equation 6.4. Lags greater than the number of observations for option series are zero. The average adjustment factor, averaging across all the option series, is then calculated for each lag. A mutual $\hat{m}$ lags to later calculate the factors are separately chosen for call and put options, where the average converges at $\hat{m}$ lags.

Definition 6.5 (Adjusted sample variance) *Then, the estimate of the adjusted sample variance is*

$$\hat{\sigma}^2 = \left[\sum_{i=1}^{n}(x_i - y_i - \hat{\mu})^2 \hat{w}_i(\hat{m})\right] \Big/ (n-1) \tag{6.5}$$

where w_i is the Newey-West adjustment factor for the option series on which i^{th} observation is based.

Remark 6.3 The average Newey-West adjustment factor nearly converges at $\hat{m} = 20$ for both call and put options on the S&P 500 index in 2017.

Definition 6.6 (t-Statistic) *The t-statistic to test whether the Diff is significantly different from zero is*

$$t\text{-}statistic = \frac{\hat{\mu}\sqrt{n}}{\hat{\sigma}} \tag{6.6}$$

Remark 6.4 The t-statistic approaches to a z-statistic because the sample size covered in this study is nearly 300,000.

6.2 Empirical results

6.2.1 Cross-validation

The optimal stopping value $\hat{M}$ in Equation 5.14 that controls the number of additive expansions of the model $\hat{f}(\mathrm{x})$ is obtained by cross-validation. The average $\hat{M}$ for the 24-month training samples are approximately equal to 11 for both call and put options on the S&P 500 index.

6.2.2 Relative importance of predictor variables

Table 6.1 presents that the location parameter of changes in the implied volatility $\Delta\sigma_{imp}$ is chosen most of the time for both call and put options on the S&P 500 index. In addition, the underlying price change variable ΔS constitutes more weights for call than put options.

In addition, 15%−25% of total chosen splits at the underlying price change ΔS for call and put options suggests that its exponentials are likely to be significant explanatory variables.

Relative importance of predictor variables

	Call options				Put options			
OS month	$\Delta\sigma_{imp}$	ΔS	T	δ_{BS}	$\Delta\sigma_{imp}$	ΔS	T	δ_{BS}
2017-01	61.6%	26.8%	7.7%	3.9%	64.4%	15.7%	12.0%	7.9%
2017-02	64.4%	24.5%	6.8%	4.3%	68.2%	14.8%	10.3%	6.7%
2017-03	64.8%	23.3%	8.2%	3.7%	67.7%	15.4%	10.5%	6.4%
2017-04	67.7%	21.2%	8.1%	2.9%	66.9%	14.7%	10.9%	7.5%
2017-05	67.6%	21.9%	7.0%	3.6%	65.6%	16.4%	11.8%	6.2%
2017-06	67.5%	21.3%	7.3%	3.8%	64.5%	16.8%	11.7%	7.1%
2017-07	68.1%	21.3%	7.4%	3.2%	66.5%	16.6%	10.8%	6.1%
2017-08	68.4%	20.6%	7.7%	3.3%	68.2%	16.4%	9.2%	6.2%
2017-09	65.3%	23.0%	7.7%	4.1%	65.9%	15.9%	11.3%	6.9%
2017-10	65.0%	23.3%	8.3%	3.4%	66.1%	15.0%	10.9%	8.0%
2017-11	64.0%	23.9%	9.2%	2.9%	65.5%	15.9%	10.0%	8.5%
2017-12	65.8%	21.4%	9.0%	3.8%	64.0%	16.6%	11.0%	8.3%
Average	65.9%	22.7%	7.9%	3.6%	66.1%	15.8%	10.9%	7.2%

Table 6.1: Relative importance of automatically chosen split variables when applying the tree-boosting algorithm for MV hedging residuals for call and put options on the S&P 500 index in 2017.

6.2.3 Gains and improvement

Hedging residuals

	Call options			Put options		
	ε_{BS}	ε_{MV}	ε_{MVB}	ε_{BS}	ε_{MV}	ε_{MVB}
count	130,413	130,413	130,413	162,681	162,681	162,681
mean	-0.000157	-0.000103	-0.000076	-0.000173	-0.000153	-0.000127
std	0.001254	0.001200	0.001198	0.000957	0.000976	0.000966
min	-0.014966	-0.014737	-0.014825	-0.017846	-0.017681	-0.017684
25%	-0.000594	-0.000529	-0.000502	-0.000466	-0.000456	-0.000429
50%	-0.000119	-0.000088	-0.000063	-0.000133	-0.000117	-0.000102
75%	0.000308	0.000340	0.000359	0.000144	0.000173	0.000191
max	0.050641	0.051572	0.051495	0.037754	0.038601	0.038696

Table 6.2: Descriptive statistics of hedging residuals by different models

Firstly, we carry out a naive analysis based common descriptive statistics in Table 6.2. Both the MV and MVB deltas result more near-zero hedging residuals with lower absolute values of means and medians. The MV residuals for put options are more dispersed than the BS ones with greater standard deviation and interquantile range. There is no improvements observed in minimum and maximum residuals. Nonetheless, the MV and MVB models improve their prior models as expected in the other cases.

Model gains over the BS model

	Gain over BS				t-statistic	
	Call		Put		Call	Put
$\lvert\delta_{BS}\rvert$	MV	MVB	MV	MVB	MVB-MV	
0.1	70.2%	69.0%	67.3%	67.3%	12.3	-4.1
0.2	70.7%	70.6%	66.9%	67.1%	-26.7	-46.7
0.3	70.6%	70.9%	66.5%	67.1%	-42.2	-67.5
0.4	70.2%	70.6%	66.1%	66.9%	-46.8	-75.9
0.5	70.3%	70.6%	65.7%	66.6%	-47.8	-75.4
0.6	71.0%	71.1%	64.8%	65.7%	-28.3	-63.0
0.7	71.8%	71.6%	64.1%	64.8%	-9.4	-51.6
0.8	72.2%	72.0%	64.7%	65.0%	3.1	-28.5
0.9	71.6%	71.4%	64.7%	65.2%	20.2	-44.1
All	71.3%	71.2%	65.4%	65.9%	-30.7	-142.7

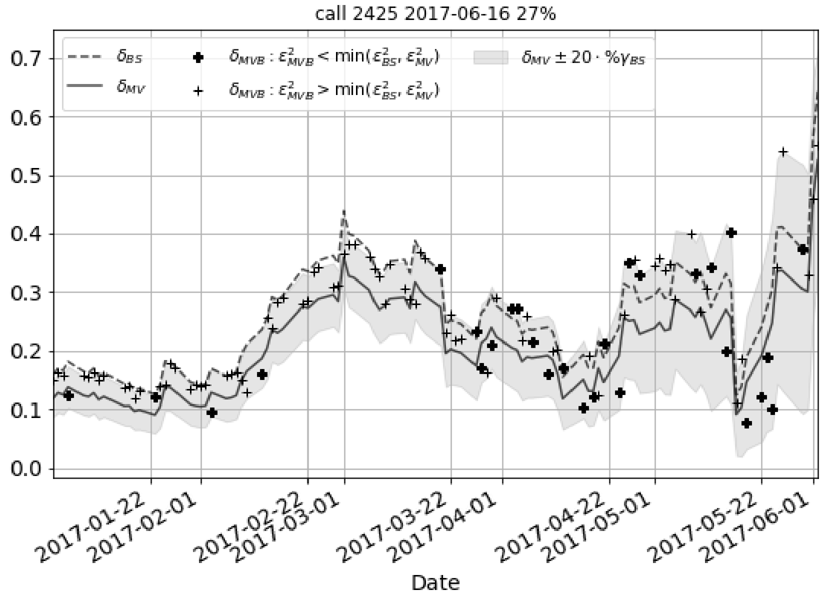
Table 6.3: Out-of-sample average hedging Gain calculated by Equation 6.1 in 2017. The MV model parameters a, b, and c in Equation 5.4 are estimated using all options traded in a 24 month window and then applied to determine the hedge on every day in the next month. Then, the proposed boosting method (MVB) further fits $\hat{f}(\mathrm{x})$ into the MV residuals as Equation 5.10 using the same 24-month periods. The boosting models are used to determine the hedge in the following test month. Gain is percentage reduction in sum of squared errors compared to the BS model. The t-statistic measures significance of difference that the MVB model has over the MV model using Newey-West adjustment factor in Equation 6.4.

Secondly, we further analyze improvement of the MVB model. Table 6.3 presents detailed results for gains obtained by each delta bucket. When the residuals for all options are included, the average Gain of the MV model is about 71.3% and 65.4% for call and put options respectively. ITM call options have higher

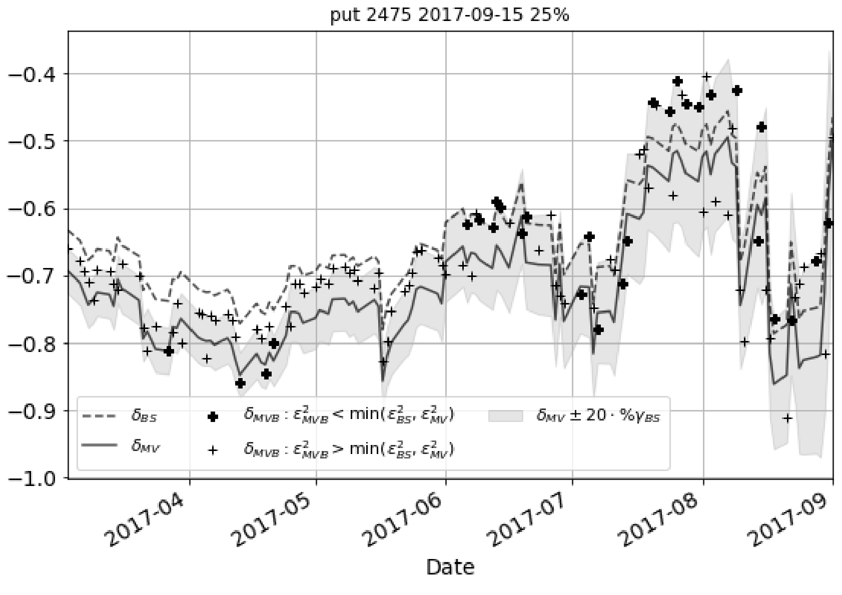
MV gains than OTM options and it is otherwise for put options. Nonetheless, differences by moneyness are insignificant.

The MVB delta statistically improves the MV delta in all delta buckets for put options with higher gains over the BS delta and their t-statistic values less than -1.645 (one tailed 95% confidence level). In contrast, it is not the case for call options. The hedging performance of the MVB delta seems even worse than the MV one for deep ITM and OTM call options.

Thirdly, the MVB delta effectively bounded by $\delta_{MV} \pm 20 \cdot \%\gamma_{BS}$ constitutes approximately 85% of observations where the MVB model results in the least squared error hedging errors. Hence, expanding this boundary range is expected to improve the MVB performance. However, this is still the trade-off for the model variance and hedging costs in practice as discussed in Section 5.3.3.



(a) Call option with strike at 2,425 and expiration at 16 June 2017



(b) Put option with strike at 2,475 and expiration at 15 September 2017

Figure 6.2: Delta ratios by different models for two options on the S%P 500 index in 2017: (a) call option with strike at 2,425 and expiration at 16 June 2017, and (b) put option with strike at 2,475 and expiration at 15 September 2017. $\delta_{MVB} : \varepsilon^2_{MVB} < \min(\varepsilon^2_{MV}, \varepsilon^2_{BS})$ is the MVB delta resulting in the least squared hedging error among the three models. The MVB delta yields the minimum ε^2 in 27% and 25% of observations for the above sample call and put options respectively.

Chapter 7

Conclusions

The assumption of constant volatility of the Black-Scholes model has inspired many studies in options pricing and hedging. Hull and White (2017) proposed an empirical model that estimates minimum-variance delta which minimizes variance of hedging errors. They considered the expected change in the implied volatility empirically as a quadratic function of the BS delta, given an underlying price change. Intuitively, the model attempted to adjust the BS hedging residuals to improve hedging performance. They found that the model reduced variance of hedging errors between 2004 and 2015 for options on the S&P 500 index.

This research proposes a model of gradient boosting algorithm using regression trees to further adjust the hedging residuals by the above model and improve hedging performance. The predictor space includes the BS delta, option time to maturity, change in underlying price, and change in the implied volatility. Those variables are essential properties of an option. Feature selection property of the boosting is supposed to allocate optimal weights to the input variables. There are boundaries for the algorithm to be practically effective.

We carry out an empirical study on the S&P 500 index options between 2015 and 2017. The quadratic equation empirically still holds for call and put options similarly to the prior study. The Hull and White (2017) model provides significant gains over the BS model. Gains for call options are higher than that for put options.

The proposed method improves the hedging gains for put options only. Change in the implied volatility is a dominantly chosen split variable in the boosting model. There are three possible reasons for the results. Firstly, the predictor space does not include enough explanatory variables for hedging residuals. Secondly, the hedging gain for call options by the Hull and White model is too high for our model to improve. The gain after the boosting for put options is still worse than the original gain for call options. Thirdly, boundaries for delta computed by the proposed method are too tight so robustness of the model is limited. Fourthly, the number of observations for put options are significantly higher than that for call options from 2015 to 2017. The boosting model for put options are more effective, which is possibly attributed to larger in-sample data.

In summary, this research thesis provides an empirical evidence that the boosting tree algorithm can further improve hedging performance of the Hull and White model on put options on the S&P 500 index in 2017. The limitations are the predictor space of only four variables and the short in-sample period. Hence, future studies could address this framework with a larger predictor space and a longer observation period. The statistical learning method could also be replaced.

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